

2082

## Bachelor of Science in Computer Science and Information Technology

Course Title: **Statistics I**

Code No: **STA164** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

### Section A

1. What do you mean by statistics? Describe briefly. The following are the marks of 30 students in Statistics. 60 54 22 36 49 84 79 56 28 43 80 50 60 56 57 46 39 57 73 65 69 48 75 65 59 73 46 44 70 22. By taking suitable class interval find Karl Pearson's coefficient of skewness interpret the results.
2. What do you understand by correlation analysis? Bradford Electric Illuminating is studying the relationship between kilowatt ■ hours (thousands) used and the number of rooms in a private single-family residence. A random sample of 5 houses yields the following. Number of rooms 11 9 13 7 10 Kilowatts ■ hours (thousands) used 9 7 12 4 8 (i) Find the correlation coefficient between number of rooms and kilowatts ■ hours used. Also Interpret the result. (ii) Determine the regression equation of kilowatts hours used on number of rooms. (iii) Estimate the kilowatts -hours used for a 8 rooms house.
3. Two continuous random variables X and Y have following pdf:  $f(x, y) = k \times y$ ,  $0 \leq y \leq 1$ ,  $0 \leq x \leq 1$ , otherwise 0. Determine (i) the value of k (ii) marginal pdf of X and Y (iii) whether X and Y are independent?

### Section B

*Attempt any Eight questions [8×5=40]*

4. Maximal static in respiratory pressure is an index of respiratory muscle strength. The following data show the measure of maximal static in respiratory for 11 cystic fibrosis patients: 115, 95, 100, 85, 90, 70, 45, 115, 40, 115, and 95. (i) Calculate five number summary. (ii) Construct a box- and ■ whisker plot. Also comment on the shape of the distribution.
5. Calculate an appropriate measure of central tendency from the following table. Give the reason for your choice of measure of central tendency. Age in years: Below 20, 20-30, 30-40, 40-50, 50 and more. Number of persons: 12, 27, 25, 15, 11. Also find the lowest age of oldest 20% persons.
6. Define independent events in probability. A problem of statistics is given to three students A, B and C whose chances of solving the problem are in the ratio 2: 3: 5. Find the probability that (i) all of them solve the problem. (ii) the problem will be solved.
7. State Bayes' theorem of probability. A computer maker receives parts from three suppliers S1, S2, and S3. Fifty percent come from S1, thirty percent from S2, and twenty percent from S3. Among all the parts supplied By S1, 5% are defective, those supplied by S2, 3% are defective and 4% respectively. A customer complains

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that a certain part in his recently purchased computer is defective. What is the probability that it was supplied by S1?

8. What is sampling? Give main objects of sampling.
9. Under what conditions the binomial distribution is used? A multiple choice test has 5 questions. There are 4 choices for each questions of which one is correct. A student who has not studied for the test decides to answer all the questions randomly. What is the probability that he will get (i) all questions correct (ii) at least four questions correct?
10. Under what conditions binomial distribution tends to poisson distribution? If the probability that an individual suffers an adverse reaction from a particular drug is known to be 0.0001, determine the probability that out of 20000 individuals (i) exactly 3 individual will suffer an adverse reaction (ii) more than one individual will suffer an adverse reaction.
11. The finished inside diameter of a piston ring is normally distributed with a mean of 10 cm and standard deviation of 0.04 cm. (i) What proportion of rings will have an inside diameter exceeding 10.08 cm (ii) Below what value of inside diameter will 10% of the piston rings fall?
12. Given the following bivariate probability distribution:

|     |     |     |     |      |     |     |      |     |     |     |
|-----|-----|-----|-----|------|-----|-----|------|-----|-----|-----|
| X\Y | 01  | 157 | 151 | 1513 | 152 | 151 | 1522 | 151 | 152 | 15  |
| 101 | 0.1 | 0.2 | 0.3 | 0.4  | 0.5 | 0.6 | 0.7  | 0.8 | 0.9 | 1.0 |

Obtain (i) marginal probability distribution of X (ii)  $P(X < 0, Y = 2)$  (iii) conditional probability distribution of X given Y = 0. OR Write short notes on any two. (i) Primary data and secondary data (ii) Kurtosis (iii) Regression analysis.